SymPy Bug Card

Bug 33: solveset Represents an Unsatisfiable Tangent Equation by Infinities

Task. Solve the complex trigonometric equation

$$\tan(x) = i, \qquad x \in \mathbb{C}.$$

SymPy's answer.

$$\mathrm{solveset}(\tan(x) = i,\ x,\ \mathbb{C}) = \mathrm{ImageSet}(\Lambda(\_n,\ \_n\pi + oo\,i), \mathbb{Z}).$$

Correct answer.

$$\varnothing.$$

Explanation. For finite z,

$$\tan(z) = -i\,\frac{\exp(2iz) - 1}{\exp(2iz) + 1}.$$

Setting $\tan(z) = i$ forces $\exp(2iz) = 0$, impossible for a finite complex argument. The returned $oo\,i$-valued image set is therefore not a finite complex solution family.

Likely root cause. In `sympy/solvers/solveset.py`, `_invert_trig_hyp_complex` constructs the tangent solution pattern $n\pi + \arctan(a)$ without rejecting non-finite inverse values; here $\arctan(i)$ evaluates to $oo\,i$. This is a new instance of a known failure mode.

Artifact (GitHub). `bug-report/bug-033-solveset-represents-an-unsatisfiable-tangent-equation-by-inf/`